

of the image which is to be used to create the container named *lxc*. The name of the image follows this syntax: `<repository>:<image name>`. So, essentially, *ubuntu* in that command is the name of the image repository. Note that the name of the container must be a valid host name.

```
root@ubuntu:~# lxc launch ubuntu: first-container
Creating first-container
Starting first-container
```

```
To get into your container,
root@ubuntu:~# lxc exec first-container /bin/bash
root@first-container:~# ifconfig
eth0      Link encap:Ethernet  HWaddr 00:16:3e:b1:5d:bb
          inet addr:10.134.32.8  Bcast:10.134.32.255
          Mask:255.255.255.0
          inet6 addr: fe80::216:3eff:feb1:5dbb/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:18 errors:0 dropped:0 overruns:0 frame:0
          TX packets:14 errors:0 dropped:0 overruns:0
          carrier:0
```

```
collisions:0 txqueuelen:1000
RX bytes:2055 (2.0 KB)  TX bytes:1837 (1.8 KB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

LXD has a lot of other features, all of which cannot be covered in one article. If you'd like to explore LXD in more depth, head on to Google and LXD's official GitHub repository. <https://github.com/lxc/lxd>

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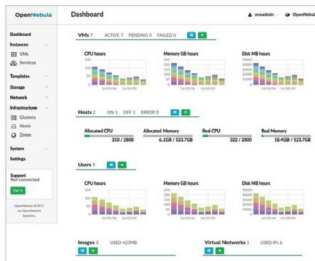


Figure 2: OpenNebula Sunstone dashboard

can be any of the following: *create*, *deploy*, *shutdown*, *livemigrate*, *migrate*, *hold*, *release*, *stop*, *cancel*, *suspend*, *resume*, *delete*, *list*, *show*, *top* and *history*. See the OpenNebula documentation for more on these commands.

The *create* command allows you to submit a new VM to the OpenNebula cluster. This command requires the user to pass the VM template file. Once you submit a VM, you can view its status with *show*. You will also know the ID assigned to the VM. To apply commands like *shutdown*, *stop*, *cancel*, *suspend* and *resume* to a VM, just specify the VM ID.

OpenNebula allows a user to migrate a VM from one host in the cluster to another. With *migrate*, the VM is first stopped before it is redeployed on another machine,

whereas *livemigrate* migrates the VM without stopping and redeploying it. To specify a migration, you need to specify both the host ID and the VM ID.

Launching a virtual machine

To launch the virtual machine, type the following commands:

```
01 nebula-user@nebula-cloud-server:~$ onenv create kvmVM.
template
02 nebula-user@nebula-cloud-server:~$ onenv list
03 ID      NAME STAT CPU    MEM    HOSTNAME    TIME
04 0      kvmVM pend  0      0      00          00:00:07
```

The private cloud set-up is complete.

OpenNebula Sunstone

OpenNebula Sunstone is a graphical user interface (GUI), intended for both end users and administrators, which simplifies the typical management operations in private and hybrid cloud infrastructures. OpenNebula Sunstone allows one to easily manage all OpenNebula resources and perform typical operations on them. You will be able to manage the virtual and physical resources just as we did with the CLI. <https://github.com/OpenNebula/one>

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